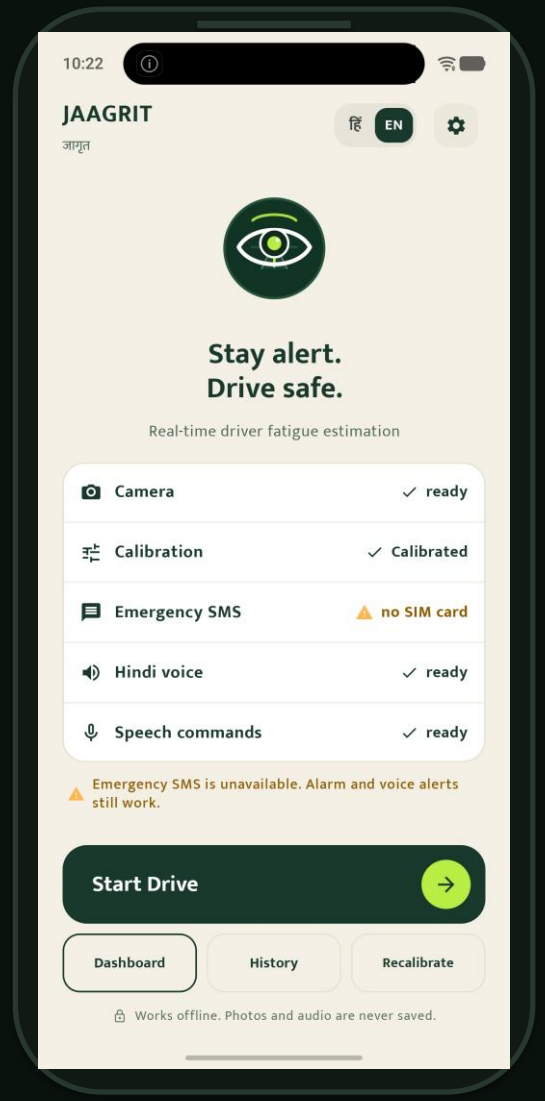
JAAGRIT जागृत

Engages before it alarms.

Completely offline.

Real-time driver alertness assistant running locally on the iQOO 15.

100% OFFLINE | 18 ms VISION | GRADUATED LADDER



THE REALITY OF THE ROAD

Drowsiness does not announce itself — it arrives in silence.

01

Micro-sleep is stealthy

Fatigue accumulates quietly over hours. A driver doesn't realize their alertness has eroded until the first involuntary 2-second eye closure.

02

Buzzers trigger too late

Traditional cab hardware screams only after a driver is asleep. The shock causes panic, road swerving, and drivers routinely unplug them.

03

Highway blind spots

Cloud-dependent fleet monitoring fails precisely when needed most: remote highway corridors with zero cellular data connectivity.

THE JAAGRIT SOLUTION

Three pillars that turn an everyday phone into a guardian.

PILLAR 1

Continuous Vision (18 ms)

Tracks Eye Aspect Ratio (EAR) and head posture at 24.8 FPS directly on the phone's CPU, without network latency or external sensors.

PILLAR 2

Graduated Ladder (L1–L5)

Engages early with friendly spoken check-ins while fatigue builds. Reserves high-decibel alarms, family audio, and SMS for true emergencies.

PILLAR 3

Zero-Cloud Privacy (0 Bytes)

100% on-device operation. No camera frames or voice recordings ever touch storage or the cloud. Tested in complete airplane mode.

GRADUATED ESCALATION

A five-stage intervention ladder that steps in gradually.

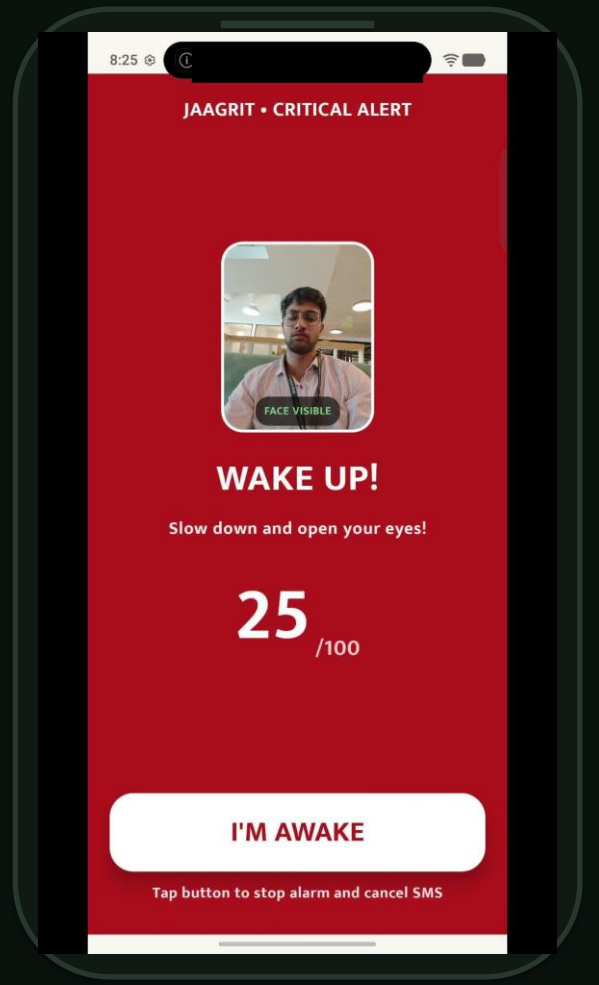
L1 **Early Warning:** Blink rate +20% or 2 yawns → Spoken friendly question (TTS Hindi)

L2 **Active Engagement:** PERCLOS ≥ 0.12 (60 s) → Spoken cognitive question; latency tracked

L3 **Direct Alert:** Eyes closed ≥ 2.5 s → Loud siren + urgent Hindi phrase + vibration

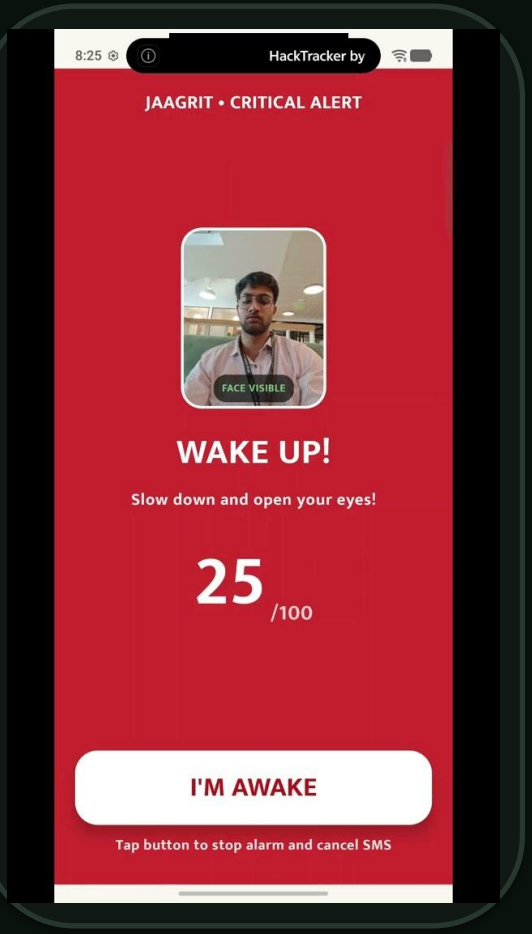
L4 **Family Voice:** No response 10 s after L3 → Pre-recorded child/family voice clip

L5 **Emergency SOS:** No response 20 s after L4 → GPS emergency SMS sent over cellular radio



LIVE PROOF & DEMO BACKUP

From subtle drowsiness to critical intervention in real time.



01. Normal Monitoring (98/100)

Green UI tracks facial landmarks continuously at 24.8 FPS. Zero CPU stutter or frame lag.

02. Drowsiness Detected (35/100)

Driver simulates eye closure. High PERCLOS (56%) immediately penalizes heuristic score into amber 'Fatigued'.

03. Critical L3 Trigger (2.5 s)

2,500 ms sustained closure immediately trips Level 3. Full-bleed red screen, alarm on USAGE_ALARM stream, and urgent Hindi TTS.

04. Instant Hands-Free Recovery

Eyes open for ≥ 3.0 s OR tapping the big 'I'M AWAKE' button silences alarm and cancels pending SMS.

PERSONAL CALIBRATION & ACCURACY

Learned from the driver's own face, not a generic average.

18 ms

Inference Latency
Snapdragon 8 Elite CPU

24.8 FPS

Camera Pipeline Rate
Continuous real-time flow

5.6x

Separation Dynamic Range
0.34 Open vs 0.06 Closed EAR

< 34°C

Sustained Phone Temp
Throttling threshold: 42°C

How Personal Calibration Works

Statistical Medians (D4): Uses medians across open/closed eyes, MAR, and response time — completely immune to single-frame outliers.

Custom Midpoint Threshold: Threshold = ClosedMedian + 0.5 × (OpenMedian − ClosedMedian). Automatically adapts to thick spectacles and narrow eye profiles.

Relative Dashboard Tilt (D18): Normalizes against the 7–9° dashboard mount tilt, so looking at road mirrors never triggers false head-droop alarms.

THE SAFETY NET (LEVEL 5)

Automated SOS with live GPS when a driver is unresponsive.

Verified SMS Template

```
ALERT: Driver may be unresponsive while driving.
Last alert: 10:37:32 AM
Alerts in this trip: 1
Location: 28.5450° N, 77.1926° E
— Sent by Jaagrit (automated safety alert)
```

Sent strictly via cellular GSM radio. Works with zero data plan or cloud server.

Failsafe Design Principles

Pre-Drive Readiness Check: The app checks cellular SIM and GPS before departure. In airplane mode, it explicitly warns the driver that SMS is offline.

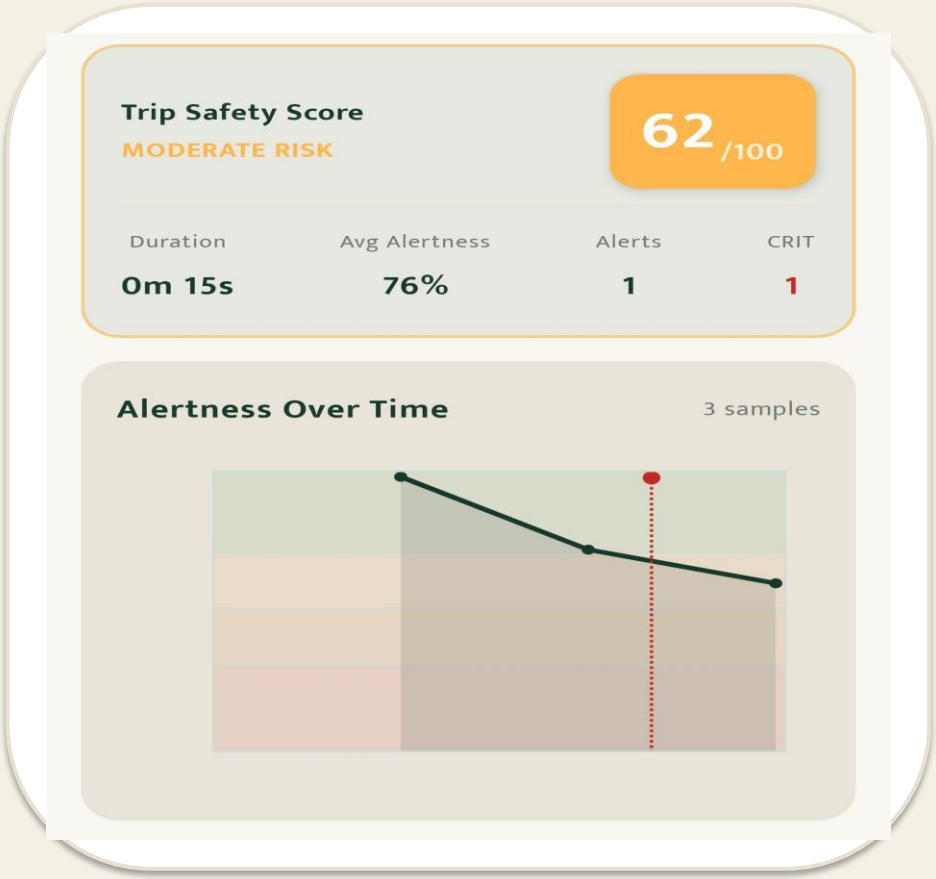
False-Positive Circuit Breaker: If >3 alerts are dismissed in 10 minutes, the app suggests recalibration rather than escalating to emergency contacts.

Any Awake Action Cancels: Tapping 'I'M AWAKE' or keeping eyes open for 3 seconds instantly halts the pending Level 5 SMS countdown.

FLEET PROOF & OFFICE KIT

Complete trip telemetry stored locally and shared seamlessly.





Office Kit Fleet Transfer

Offline Database: On-device Room database logs 48 real trips with alertness telemetry sampled every 5 seconds.

PDF Report Export: One-page visual PDF generated via Android PdfDocument and saved to Download/Jaagrit/.

Instant Laptop Mirror: Fleet managers mirror the dashboard and transfer reports in seconds via Office Kit — zero cloud portal required.

COGNITIVE PREVENTION (L1/L2)

Engaging the driver's brain before their eyes close.

Level 1: Spoken Openers

Spoken Hindi check-in:

"भाई, बहुत देर हो गई। घर पे कौन इंतज़ार कर रहा है?"

- Natural conversational tone under 8 seconds
- 60-second cooldown between prompts
- 2 ignored openers → 5-minute polite backoff
- "चुप रहो" silences the companion for 2 minutes

Level 2: Cognitive Latency Test

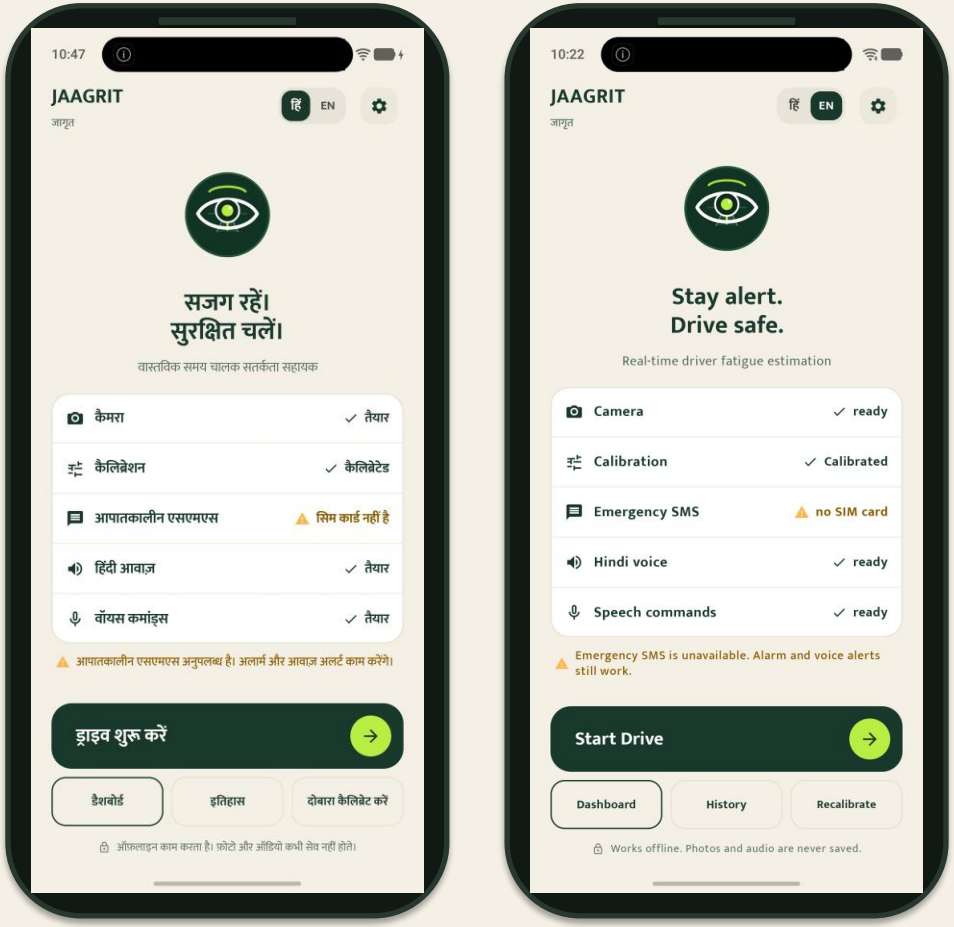
Spoken mental check:

"सौ में से तीस घटा। कितना बचा?"

- Measures response latency against calibration
- Latency > 2x baseline confirms creeping fatigue
- Microphone locked during active alarms (L3+)
- 4 fallback touch buttons ensure zero distraction

BILINGUAL & PRIVATE

Designed for Indian drivers: Hindi first, zero cloud compromise.



Zero-Cloud Guarantee

INTERNET Stripped in Manifest: android.permission.INTERNET is removed with tools:node="remove" — proven by automated security tests.

Offline Hindi Voice: Uses Google Android offline TTS Hindi engine installed locally. Zero network queries.

No Face Storage: Camera frames are analyzed in ephemeral RAM and instantly discarded. Zero image files on disk.

No Account Required: No driver sign-up, phone binding, or fleet surveillance subscriptions.

TECHNICAL DEPTH & AUDIT

Engineered for resilience: clean architecture and verified audit fixes.

CameraX
KEEP_ONLY_LATEST
Front Camera

MediaPipe
FaceLandmarker
18 ms on CPU

FatigueEngine
Pure Kotlin
Zero Android APIs

Ladder
5-Level Logic
Injected Clock

Action Handlers
TTS / Siren /
SMS / Office PDF

Independent Code Audit (AUDIT-001)

- Vulnerability Found: Mixed time bases between system uptime (frames) and epoch time (engine ticks) caused timing miscalculations.
- Fix Implemented: Injected a single monotonic Clock (elapsedRealtime) across the entire engine.
- Verified with 8 dedicated regression unit tests in AuditRegressionTest.kt.

198 Unit Tests • 0 Failures

- 27 Test Suites covering vision, engine, ladder, SMS, and reports.
- Pure Kotlin engine: runs anywhere without Android emulator dependencies.
- Pipeline UI-Independence: Vision engine is decoupled from Compose Activity, ready to run in an Android Foreground Service.

TRANSPARENCY & ROADMAP

Engineering honesty: operational boundaries and what's next.

Honest Operational Limits

Low Ambient Light: Relies on phone screen at max brightness (UI-4) as fill light; pitch-black truck cabins require ambient cabin lighting.

Dark Sunglasses: Obscures eye landmarks; system gracefully shifts primary reliance to head-droop angle and vocal checkpoints.

Extreme Head Angles: Tracking degrades beyond 30° yaw/roll.

Medical Disclaimer: Jaagrit estimates alertness heuristically — it does not claim medical diagnosis or guarantee accident immunity.

The Roadmap Ahead

Android Foreground Service: Move the decoupled vision pipeline to a background foreground service to support split-screen Google Maps navigation.

Dual-Camera Support: Integrate rear camera for road-lane departure detection alongside driver facial monitoring.

Two-Way Office Kit Depot Sync: Automatic check-in report push to depot dispatcher laptops over local Wi-Fi pairing.

JAAGRIT / जागृत

Stay alert. Drive safe.

Commercial drivers are the backbone of India's economy.

They deserve protection on the phone they already own.

18 ms Vision Engine • 5-Stage Graduated Ladder • 100% Offline & Private

OPEN REPOSITORY & LIVE PHONE DEMO

github.com/eka1357/Jaagrit

iQOO 15 (Snapdragon 8 Elite) • Direct APK • Office Kit Mirror Ready

Q&A DEFENSE

Appendix: Detection accuracy under hostile edge conditions.

Spectacles & Glare: Tested with clear glasses. Isotropic pixel-space EAR preserves a 5.6x separation gap. Personal calibration absorbs lens refractions into the driver's custom baseline.

Night & Dim Cabins: The app enforces maximum display brightness (UI-4) and keeps the screen on, utilizing the phone display as an active fill-light for the driver's face.

Road Vibration & Shaking: MediaPipe FaceLandmarker operates at 24.8 FPS in live-stream mode with face-loss glitch tolerance of 300 ms, preventing momentary bumps from corrupting closure timers.

Mirror Glances vs True Nods: Head-pitch droop threshold is calculated relative to calibrated neutral dashboard tilt (7–9°), avoiding false triggers during routine mirror checks.

Q&A DEFENSE

Appendix: Why an uncompromising on-device architecture?

Deterministic Sub-500 ms Latency: Highway safety cannot tolerate cloud API round-trips. Analyzing frames locally on the Snapdragon 8 Elite guarantees instant 18 ms decisions.

Highway Cellular Blackspots: Thousands of kilometers of Indian national highways lack reliable 4G/5G data. A safety app that stops working in tunnels or ghats is lethal.

Zero Biometric Liability: Drivers fiercely resist continuous video streaming into employer cloud dashboards. Keeping all computations ephemeral in RAM earns driver adoption.

Zero Cloud Infrastructure Cost: Scales infinitely across millions of commercial vehicles without running massive GPU inference clusters or monthly cloud bills.

Q&A DEFENSE

Appendix: Defending against driver distraction and cognitive load.

Silence During Normal Driving: The companion never speaks during normal alert driving (L0). It intervenes strictly when biological fatigue metrics cross cautionary bands.

Strict Microphone Isolation During Sirens: During Level 3, 4, and 5 alarms, the microphone is blocked. This eliminates acoustic siren echo and driver speech recognition failures.

One-Word Immediate Silencing: The driver can speak "chup raho" or tap anywhere to silence conversational check-ins for 2 minutes without disabling passive eye monitoring.

Glanceable Minimal UI: The driver UI features only a high-contrast state numeral and color word. No telemetry graphs or complex dials compete for eye gaze.

Q&A DEFENSE

Appendix: Adapting for two-wheelers, helmets, and vehicle types.

Commercial Cabs, Buses & Trucks (Primary)

Optimal environment: phone sits firmly in a dashboard cradle 40–80 cm from the driver at windshield height, capturing full facial mesh.

Two-Wheeler Visor Challenges

Full-face helmets with tinted visors block eye landmark extraction. Jaagrit's architecture falls back gracefully to head posture pitch and vocal response latency.

Extreme Road Vibrations

Motorcycle handlebar mounts impart severe vibrations; our frame glitch filter ignores brief landmark drops (<300 ms) to prevent accidental triggers.

Q&A DEFENSE

Appendix: Graceful degradation without SIM or cellular signal.

Pre-Drive Readiness Honesty

Live screenshot proof: When no SIM card is present or airplane mode is enabled, the Home screen checklist flags 'no SIM card' with clear yellow status.

Core Safety Functions Remain 100% Active

Emergency SMS is strictly an auxiliary Level 5 backup. Levels 1 through 4 (TTS voice, generated siren tone, family clip, vibration) operate entirely offline without radio.

Crash-Proof Telephony Layer

The SMS dispatcher handles missing permissions, airplane mode, and carrier transmission failures gracefully without throwing unhandled exceptions.